

URBAN SYSTEMS / CASE 07

# VERTICAL CITY RETROFIT

**The next city is not built from scratch.  
It is upgraded in place.**

A strategy for aging high-rise housing  
envelope · energy · shared space · care

CONCEPTUAL HERO VISUAL / NOT A REAL PROJECT

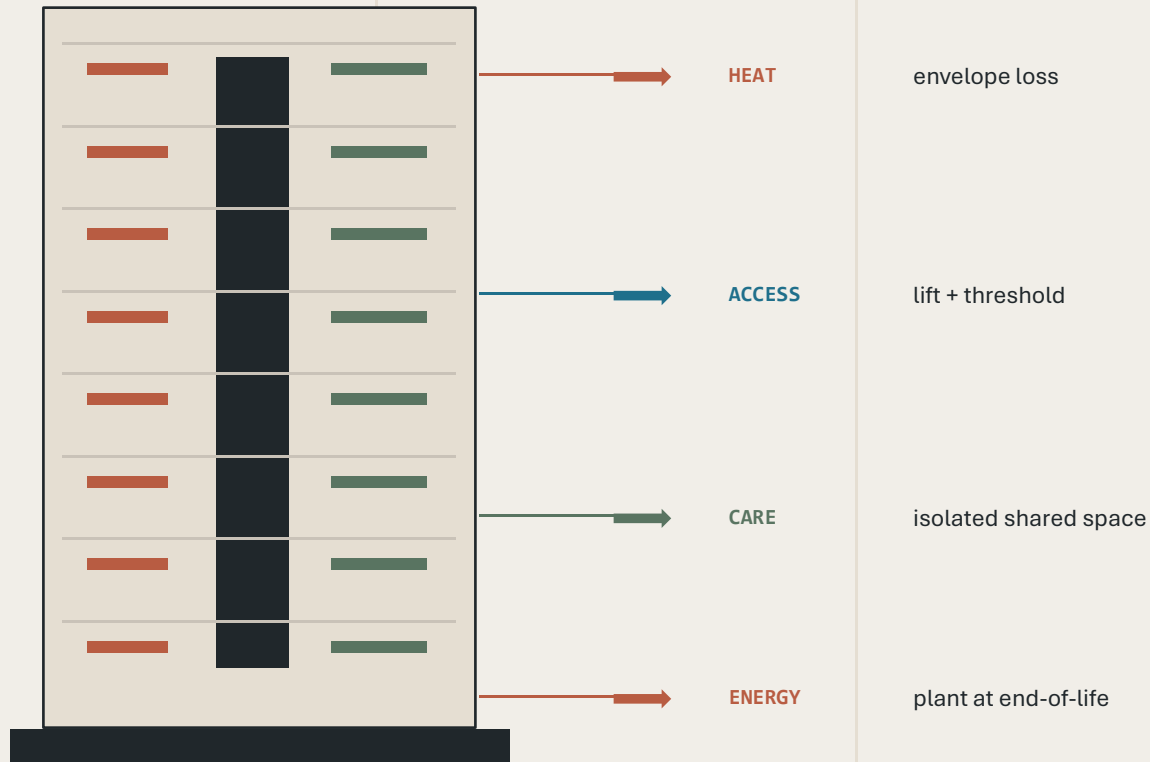
VERTICAL CITY RETROFIT · 2026



# An aging tower is a stack of hidden pressures

The façade is only the visible layer; the retrofit brief begins where energy, access, and care intersect.

## SECTIONAL DIAGNOSIS / EXISTING CONDITION

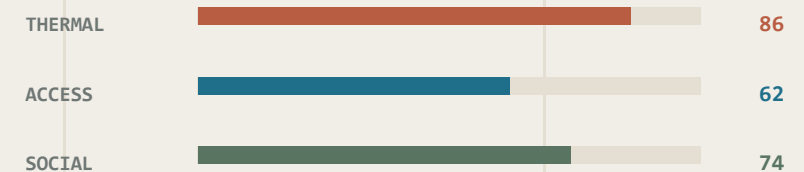


## THE RETROFIT BRIEF

### Upgrade the whole section.

A façade-only fix can lower heat loss, but it cannot repair the social and operational systems around the home.

## PRESSURE INDEX / ILLUSTRATIVE



The unit of change is the building section.

# One tower, five systems

The retrofit is not a façade package. It is a coordinated intervention across the whole vertical section.

## EXPLODED AXONOMETRIC / OPERATING SECTION

### 01 ENVELOPE

thermal skin

### 02 ENERGY

plant + distribution

### 03 MOBILITY

lift + threshold

## FROM BUILDING TO OPERATING MODEL

# Value appears in the connections.

The same section can reduce loss,  
restore access, and create care capacity.

### 04 SHARED SPACE

lobby + terrace

### 05 CARE

neighbourhood support

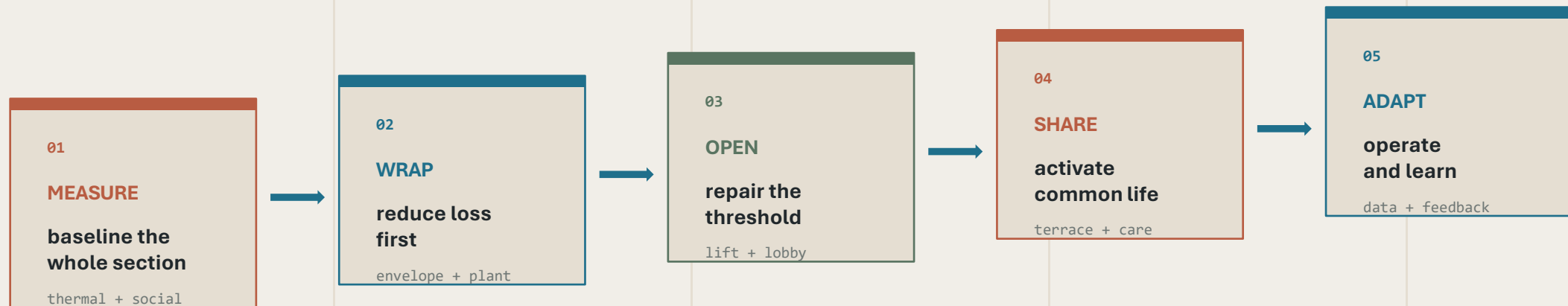
ONE SECTION / FIVE OPERATING  
LAYERS

The tower behaves like a city when its layers  
work together.

# The intervention moves in five deliberate steps

A retrofit succeeds when each move unlocks the next one without interrupting the life of the building.

## OCCUPIED-BUILDING SEQUENCE



## THE DELIVERY TEST

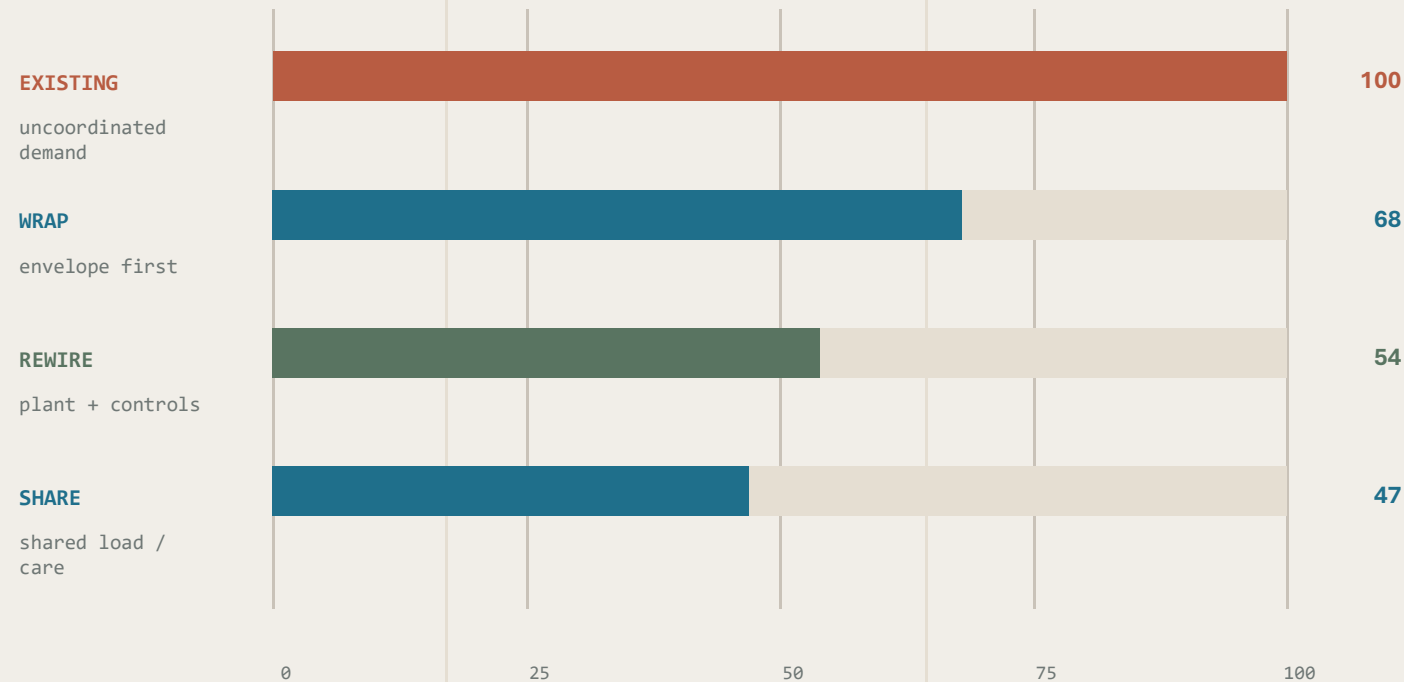
Can the building become more comfortable, more connected, and more useful while people remain inside it?

The sequence is the value proposition: reduce loss → restore access → create shared capacity → learn through operation.

# Energy is the first constraint

Before adding new capacity, reduce the amount of energy the building needs to perform everyday life.

ILLUSTRATIVE LOAD INDEX / EXISTING = 100



Illustrative sequence: reduce loss → rewire capacity → share the gain.

## THE STRATEGIC READ

### Reduce loss before adding capacity.

A plant upgrade can improve efficiency; an envelope upgrade changes the demand itself.

|    |            |                      |
|----|------------|----------------------|
| 01 | DEMAND     | lower the peak       |
| 02 | CAPACITY   | right-size the plant |
| 03 | RESILIENCE | keep options open    |

The first retrofit is a demand decision.

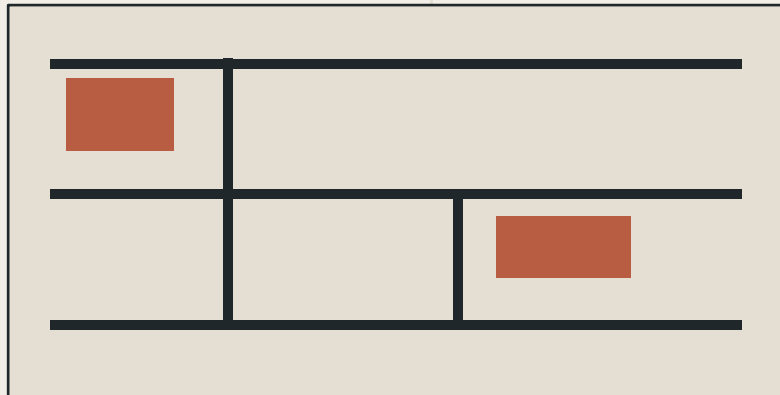
# The ground floor is the social switch

A small change at the threshold can turn circulation space into shared civic capacity.

GROUND-FLOOR PLAN / BEFORE → AFTER

## 01 / EXISTING

### sealed threshold

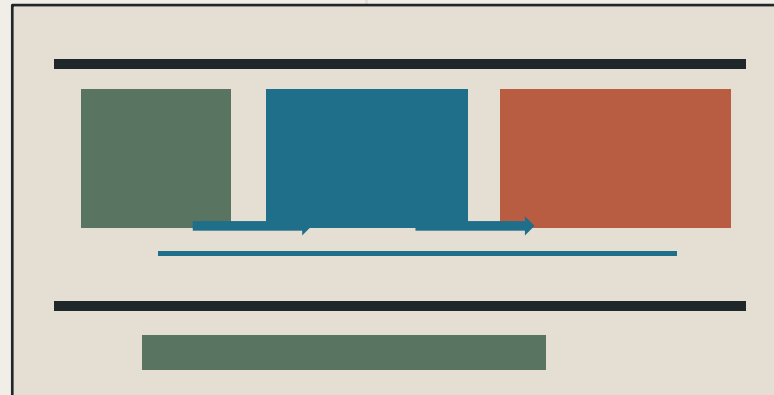


lobby / parking edge / no shared program

**Movement passes through. Nothing stays.**

## 02 / SHARED GROUND

### porous threshold



lobby / care / terrace / flexible room

**Movement meets. Something can happen.**

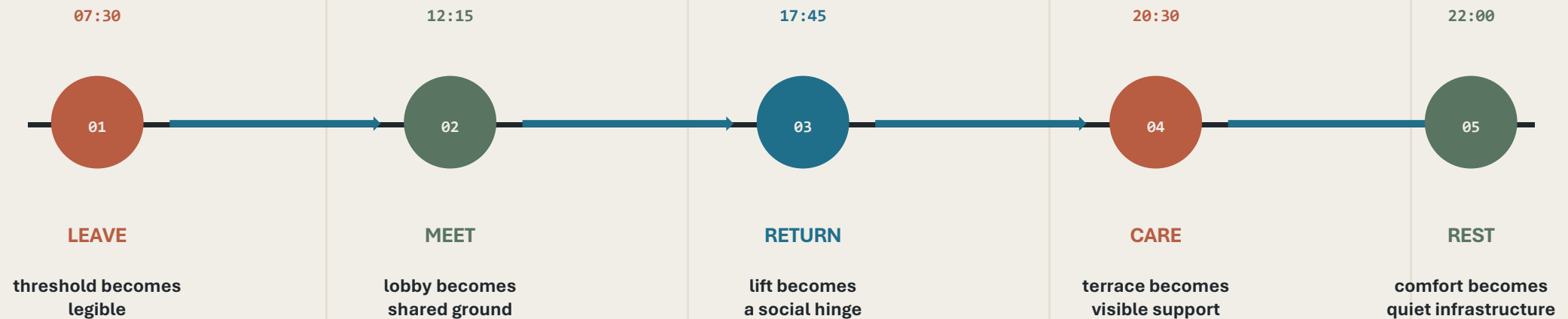
**The retrofit does not only improve the envelope; it increases the number of useful encounters the building can support.**

# A vertical city is measured in moments

The retrofit succeeds when the building supports more useful moments across an ordinary day.

DAY STORYBOARD / OCCUPIED BUILDING

FROM MOVEMENT TO CAPACITY

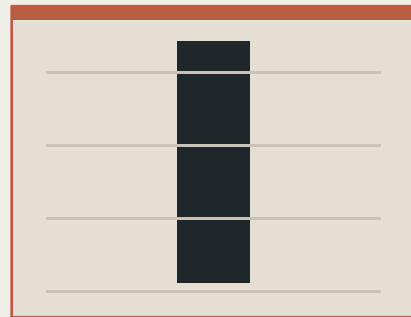


A retrofit is not finished when the construction ends; it is finished when daily life has more room to work.

# There is no single retrofit

The right depth depends on capital, disruption, and the amount of shared capacity the building needs to recover.

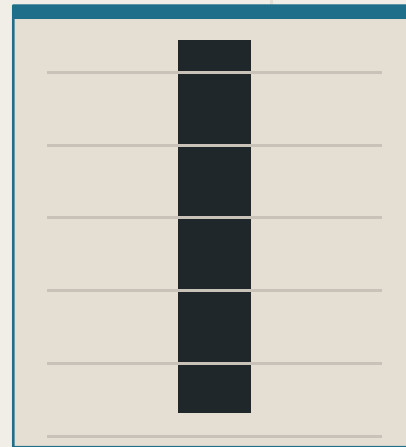
## THREE INTERVENTION DEPTHS



**LIGHT**

**protect**

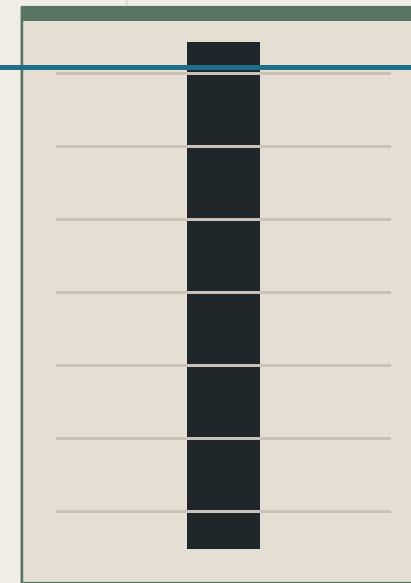
skin + controls



**DEEP**

**reconnect**

skin + threshold + care



**DISTRICT**

**share**

building + neighbourhood

## SELECTION LOGIC

**Depth follows  
shared need.**

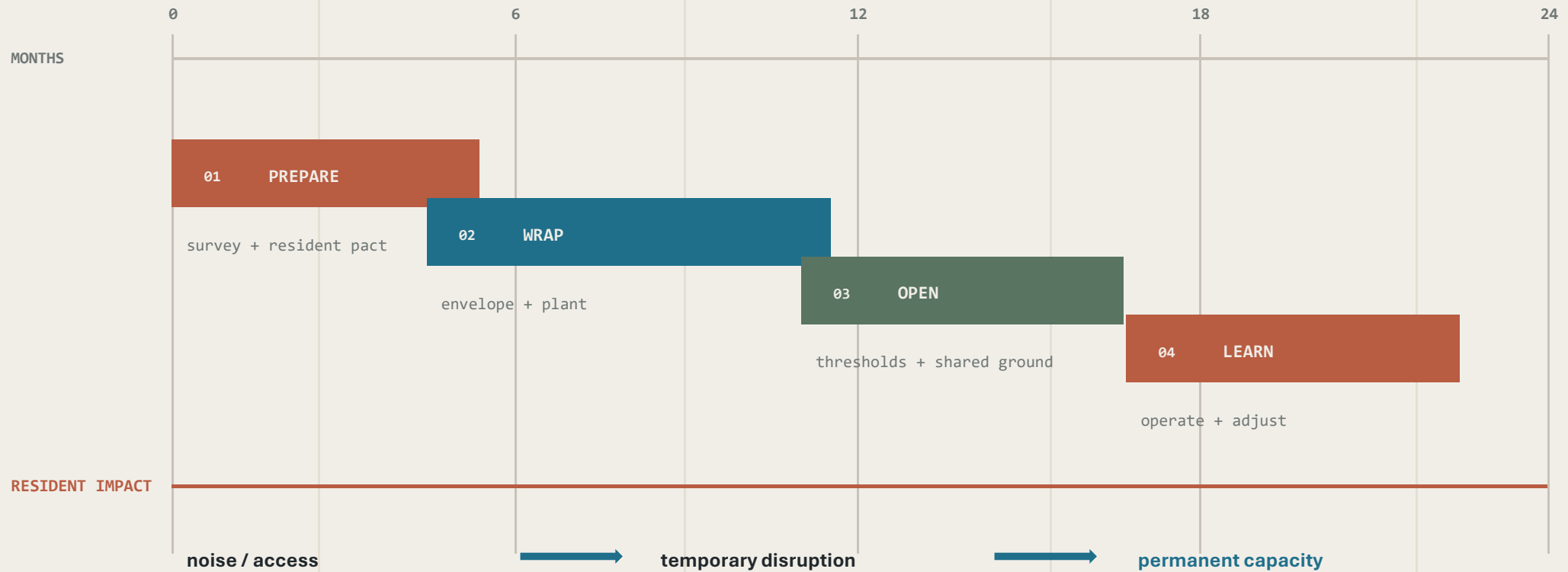
Light protects.  
Deep reconnects.  
District shares.

**Choose the smallest  
intervention that changes the  
operating model.**

# The building stays occupied while the work moves

Construction phasing is part of the design: every intervention must have a resident-safe next move.

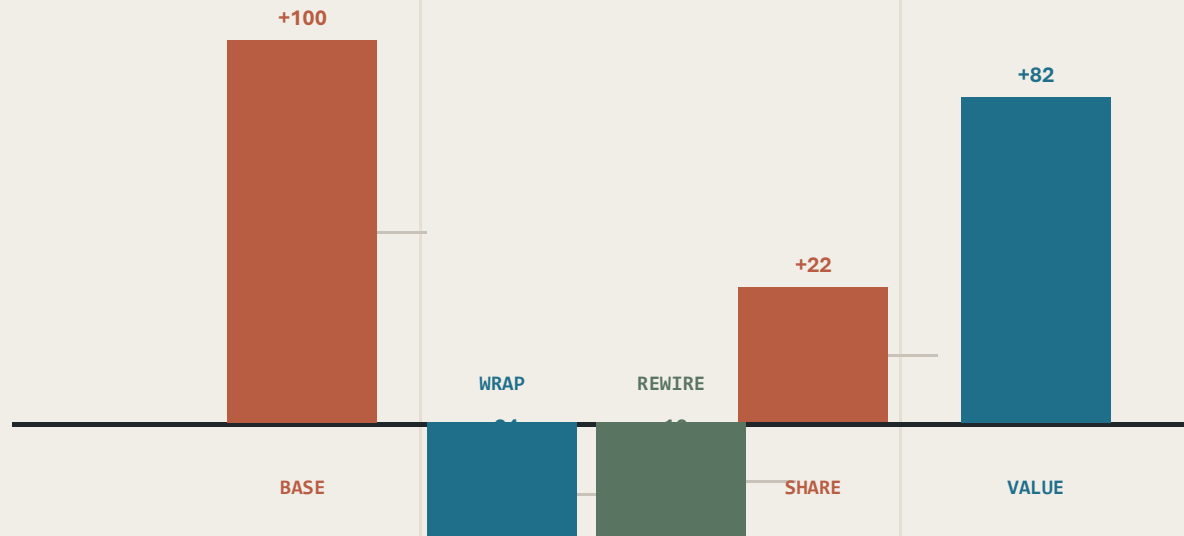
## RESIDENT-IN-PLACE PHASING



# Sequence changes the economics

The same interventions can produce different value depending on what happens first.

## ILLUSTRATIVE VALUE BRIDGE / INDEX



Illustrative index: demand reduction creates the headroom for shared capacity.

## THE ECONOMIC READ

### Order is a financial lever.

Reduce peak demand.  
Right-size capacity.  
Invest in shared use.

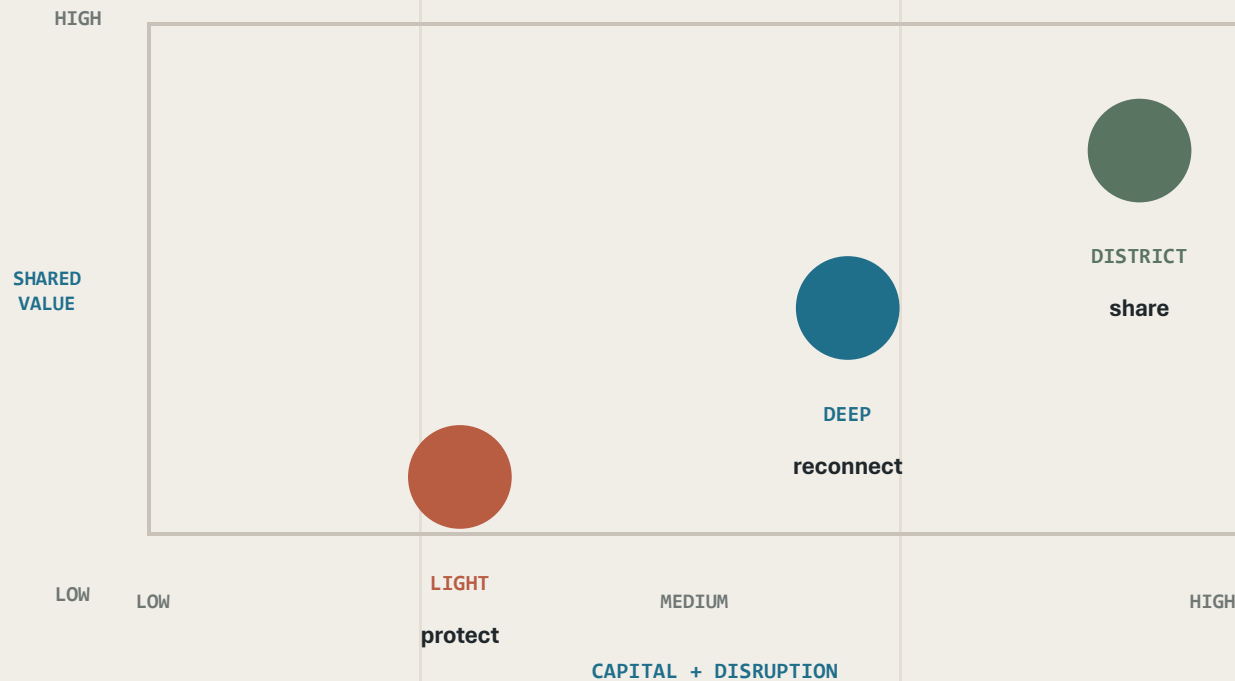
## SENSITIVITY / WHAT MOVES FIRST

|              |        |
|--------------|--------|
| CAPEX        | high   |
| DISRUPTION   | medium |
| SHARED VALUE | later  |

# Choose by trade-off, not by score

A credible retrofit brief makes the conflicts visible before it recommends a depth.

## OPTION MATRIX / STRATEGIC FIT



## THE DECISION RULE

**Pick the option that changes the system.**

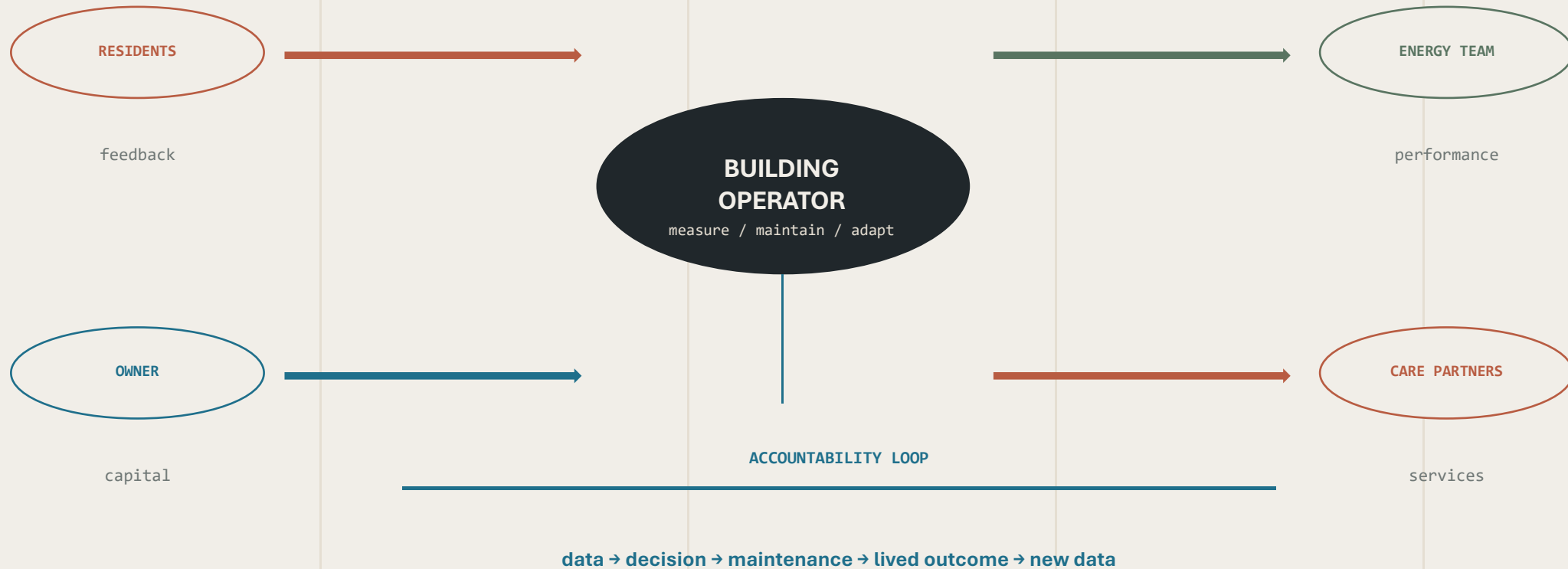
Light: minimum disruption  
Deep: balanced value  
District: maximum capacity

**Trade-offs are not a weakness; they are the decision.**

# The retrofit needs an operator

Capital can upgrade the building once; an operating model keeps the new capacity useful.

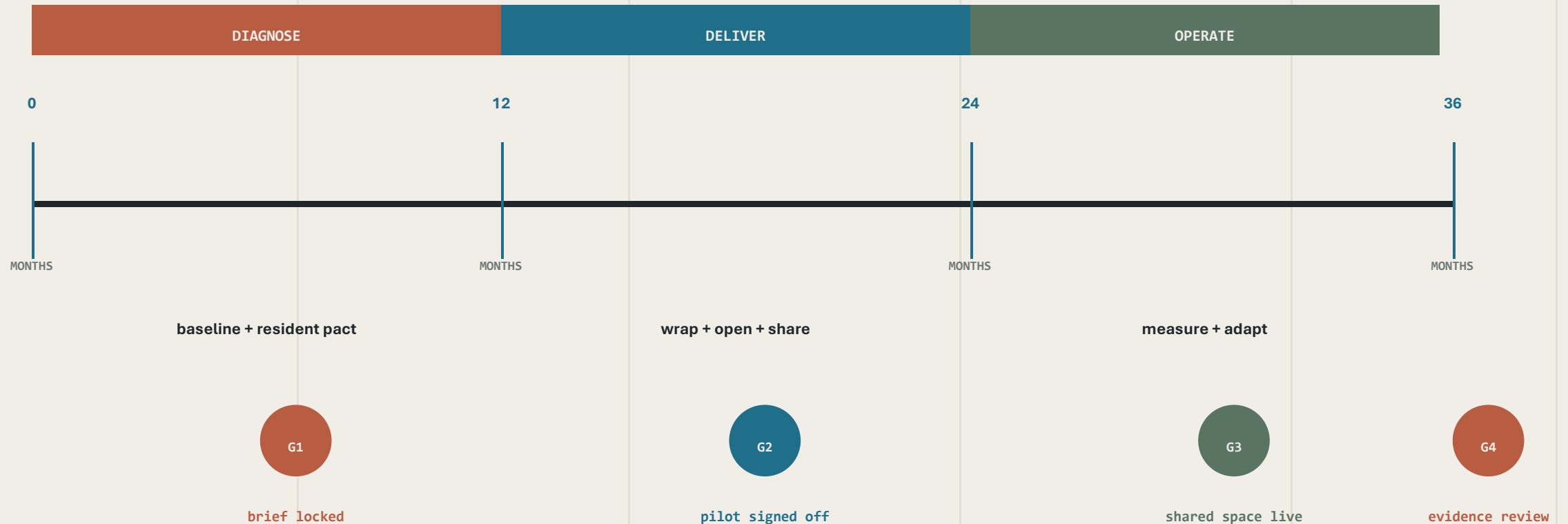
ACTORS / FLOWS / ACCOUNTABILITY



# A 36-month path with evidence at every gate

The programme becomes investable when each horizon has a decision, an owner, and proof of progress.

## THREE HORIZONS / FOUR GATES



**A gate is not a date. It is a decision supported by evidence.**

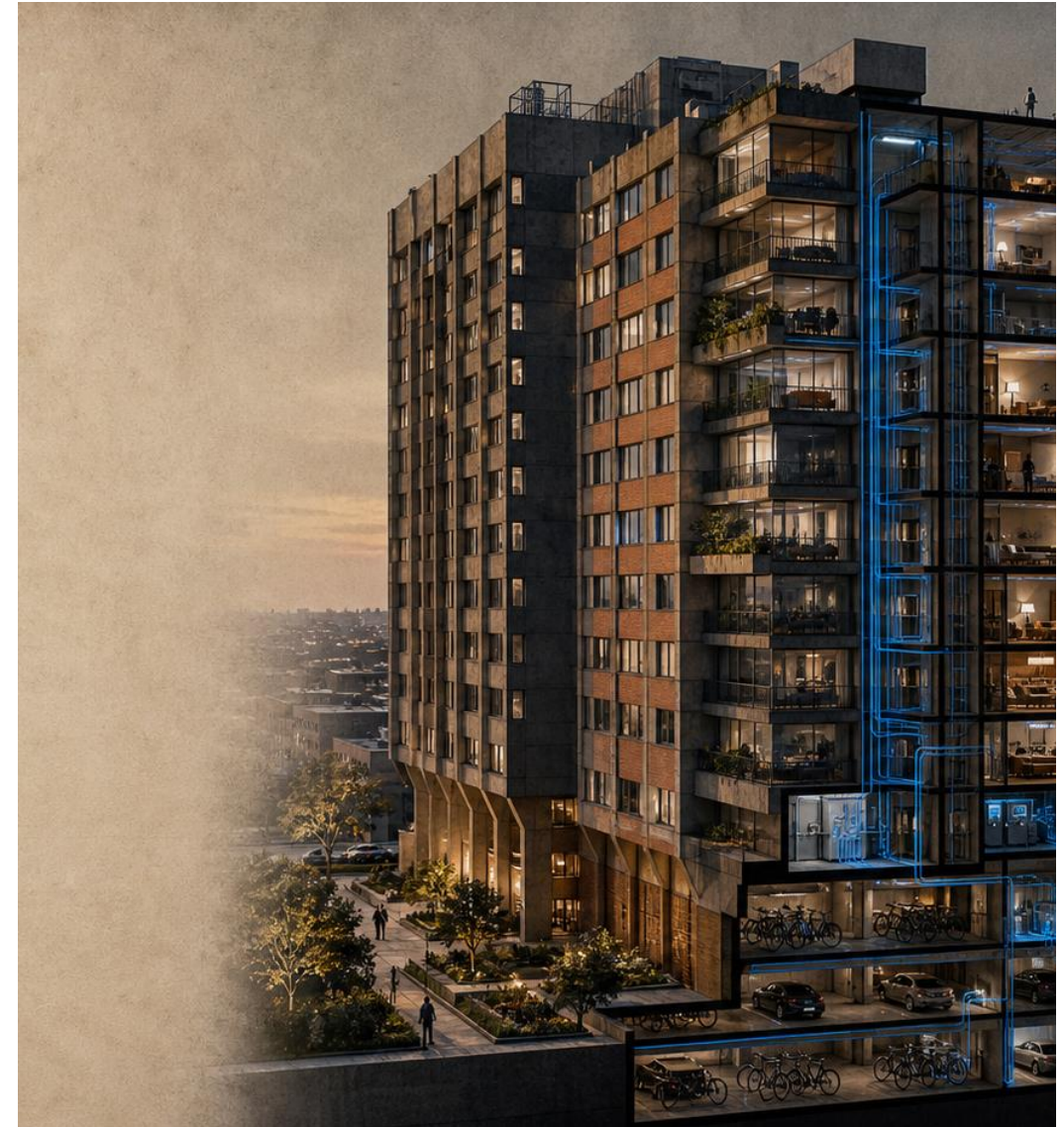
Conceptual 36-month path; milestones and gates are illustrative programme design assumptions.

# Upgrade in place. Keep life in place.

A retrofit becomes strategic when the building is treated as a connected vertical city—not a façade project.

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- 01 **Reduce loss before adding capacity.**
- 02 **Design the shared ground.**
- 03 **Operate the new capacity.**



CONCEPTUAL HERO VISUAL / NOT A REAL PROJECT